



Course on Knowledge Graphs in the era of Large Language Models

Coursework Project (Part 1): Ontology Modelling

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1 Ontology Modelling

A **knowledge scientist** is in charge of adding context to the data to make it more useful, accessible, clean, reliable and ready to be used by downstream analytics or AI-based systems. In this Part 1 of the coursework you will adopt the role of a knowledge scientist to create an ontology that formalises the domain of a selected data set in tabular format (*e.g.*, CSV file).

1.1 Dataset

You need to look for an interesting dataset (*e.g.*, in kaggle) that can use as inspiration to create the ontology in this coursework.

Example dataset. In GitHub (lab session 2) there is a dataset based on the kaggle dataset about *Pizza Restaurants and the Pizza They Sell*¹ and a model ontology covering the knowledge of that dataset.

Tips about the dataset to be selected.

- **Selection:** Select a dataset that is interesting for you, keeping in mind you will need to create an ontology about the dataset.
- **Focus:** The data may contain many things, so no need to be exhaustive. Some things can be ignored, and you may also decide to focus on some specific parts.
- **Abstraction:** The ontology needs to abstract the domain so in principle instances are not required. A few as an example/proof-of-concept may help; but it is not required to "manually" create an instance for each occurrence of a city, for example, if there is a column about locations. Other categorical information can be modelled as classes (*e.g.*, types/categories of restaurant).
- **Modelling choices:** In some cases there may be a modelling choice to add important elements as instances or subclasses. But at this abstraction stage and given problem, a hierarchy of classes is preferred.

1.2 Modelling Tasks

The first task is the selection of a dataset that is rich enough to lead to the creation of an interesting ontology. I can provide feedback on the selection.

Subtask OWL.1: Select a dataset (CSV) file about a domain of your interest. We will call this dataset `cw_data`. Motivate the selection and sketch a graph representation of the dataset (*i.e.*, abstract visual design of the domain of the dataset). This can be done on paper or assisted by a tool.

¹Kaggle dataset about pizza restaurants: <https://www.kaggle.com/datafiniti/pizza-restaurants-and-the-pizza-they-sell>. This kaggle dataset is a subset of a dataset provided by Datafiniti's Business Database.

Create a **small ontology** that models the domain of `cw_data`. Use Protégé to develop the ontology. The ontology should abstract knowledge about the domain, valid for the data at hand but potentially valid for other datasets (*e.g.*, tables).

Subtask OWL.2: Change namespace for the ontology to `http://www.uji.es/v`
`algrai/kg/2026/your_name/`.

Subtask OWL.3: Create a prefix (*e.g.*, your initials `ejr:`) for the defined namespace.

Subtask OWL.4: Reuse the annotation property `dc:creator` and annotate/indicate the ontology has been created by you.

Subtask OWL.5: Create classes and organize them into a hierarchy.

- Tip: between 15 and 25 classes.

Subtask OWL.6: Create properties and property restrictions as necessary.

- Local scope: create existential and universal restrictions (*e.g.*, around 5).
- Global scope: domain and ranges (*e.g.*, around 5).
- Create appropriate property characteristics.

Subtask OWL.7: Create annotations (*e.g.*, labels or synonyms).

Subtask OWL.8: Save the ontology into `turtle` format (*e.g.*, `.ttl`).